



**Politecnico
di Torino**

LM Ingegneria del Cinema e dei Mezzi di Comunicazione

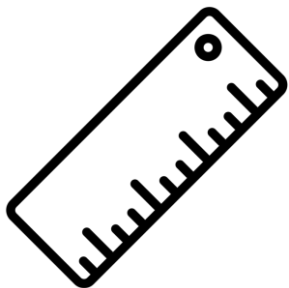
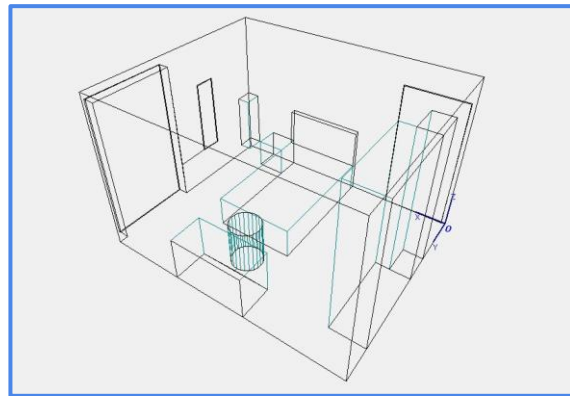
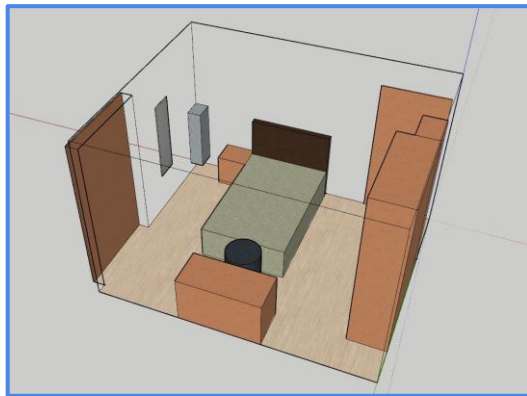
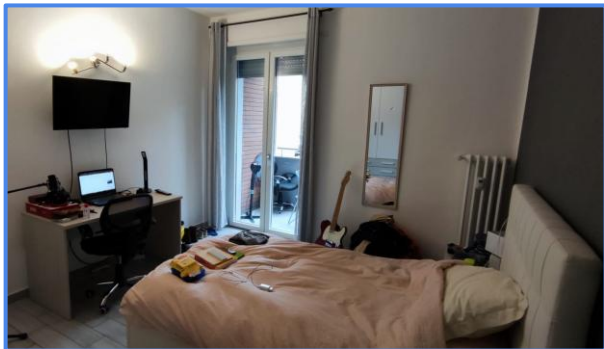
Ingegneria del Suono

A.A. 2020/2021

Room Entertainment

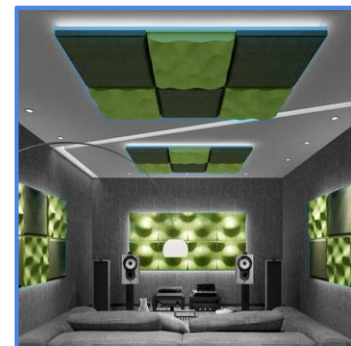
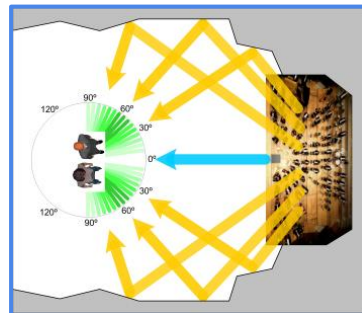
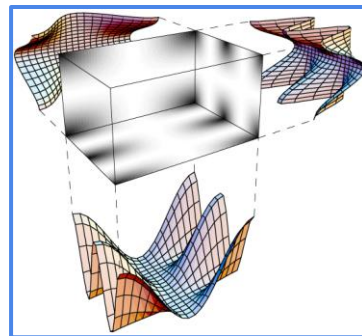
Simulazione e progetto con ODEON Combined

Stefano Calzolari S290904



Principi progettuali

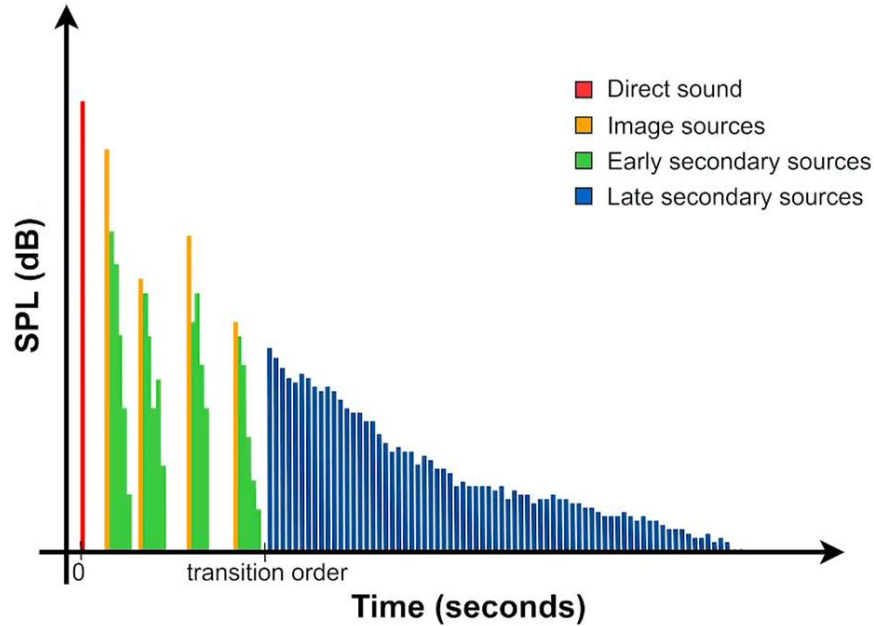
- Distribuzione modi di risonanza
- Riverbero e materiali fonoassorbenti
- Riflessioni laterali e diffusione
- Riflessioni dall'alto ed assorbimento



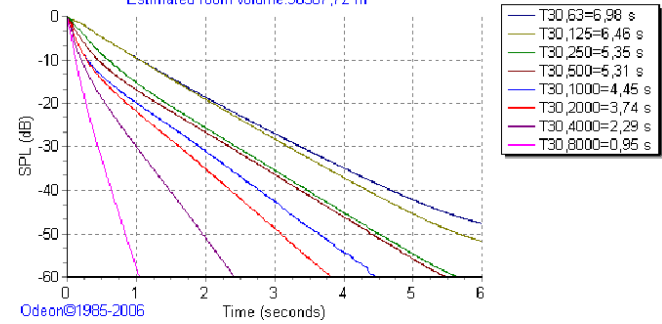
ODEON

Analisi degli algoritmi principali

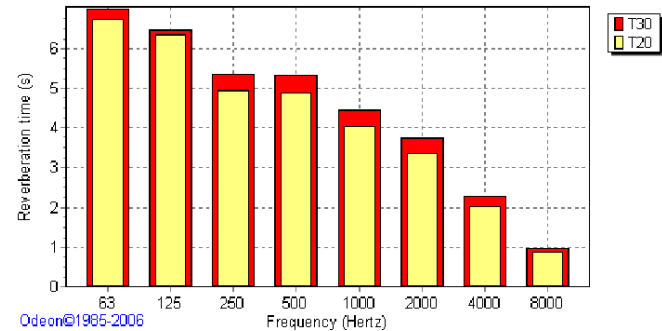
Output



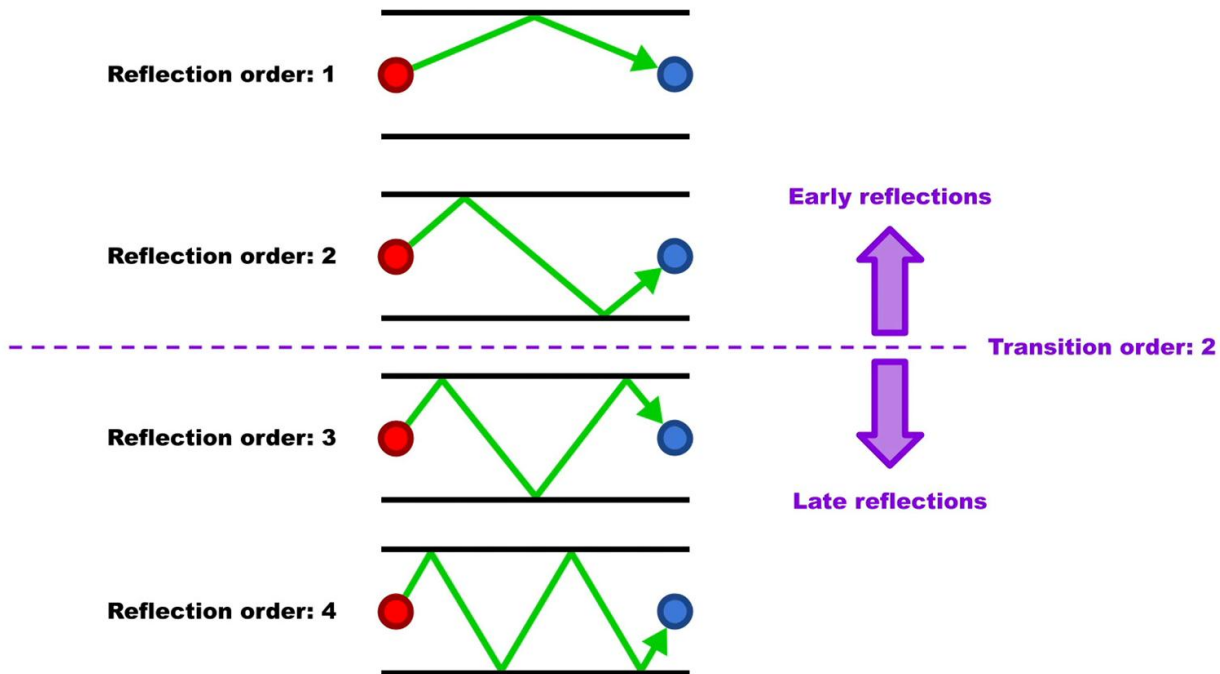
Estimated global reverberation times (Source 43, 85743 rays used)
Estimated room volume: 96387.72 m³



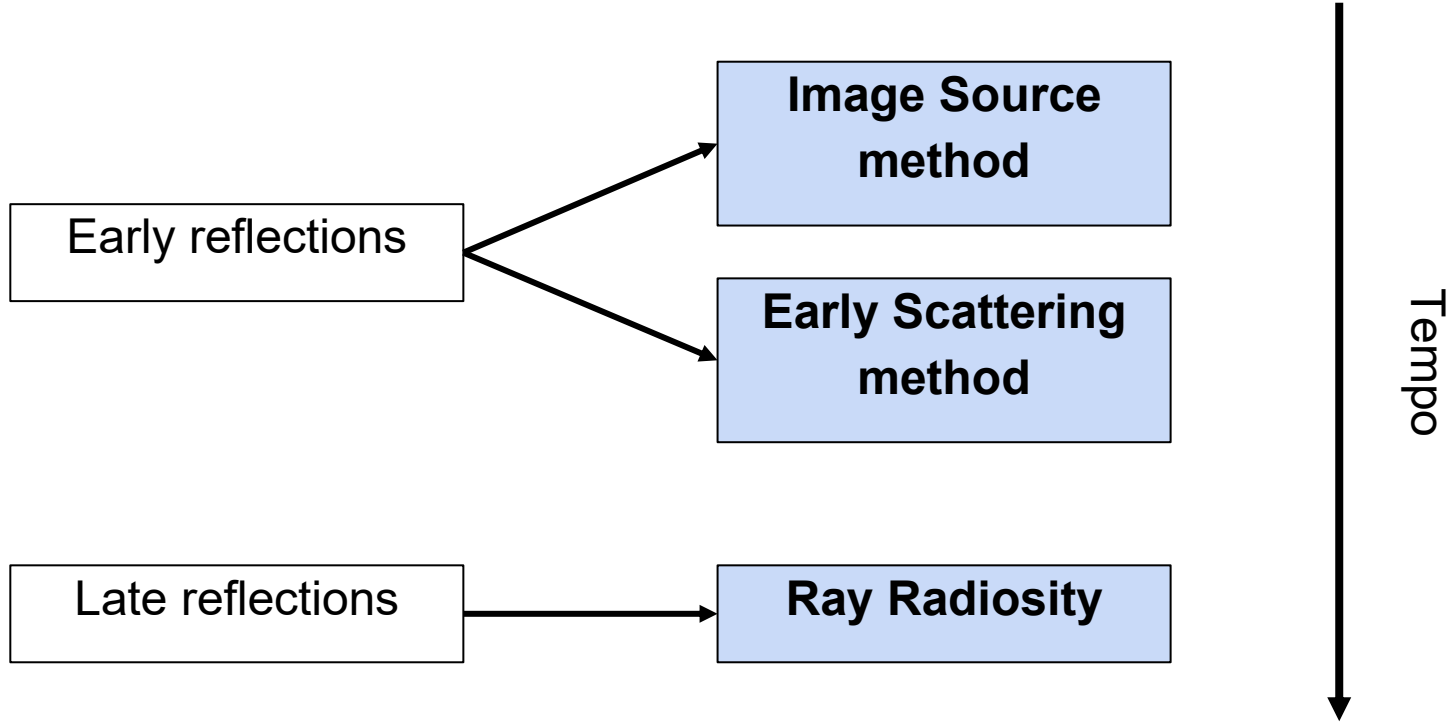
Estimated global reverberation times (Source 43, 85743 rays used)



Acoustic Rays



Algoritmi



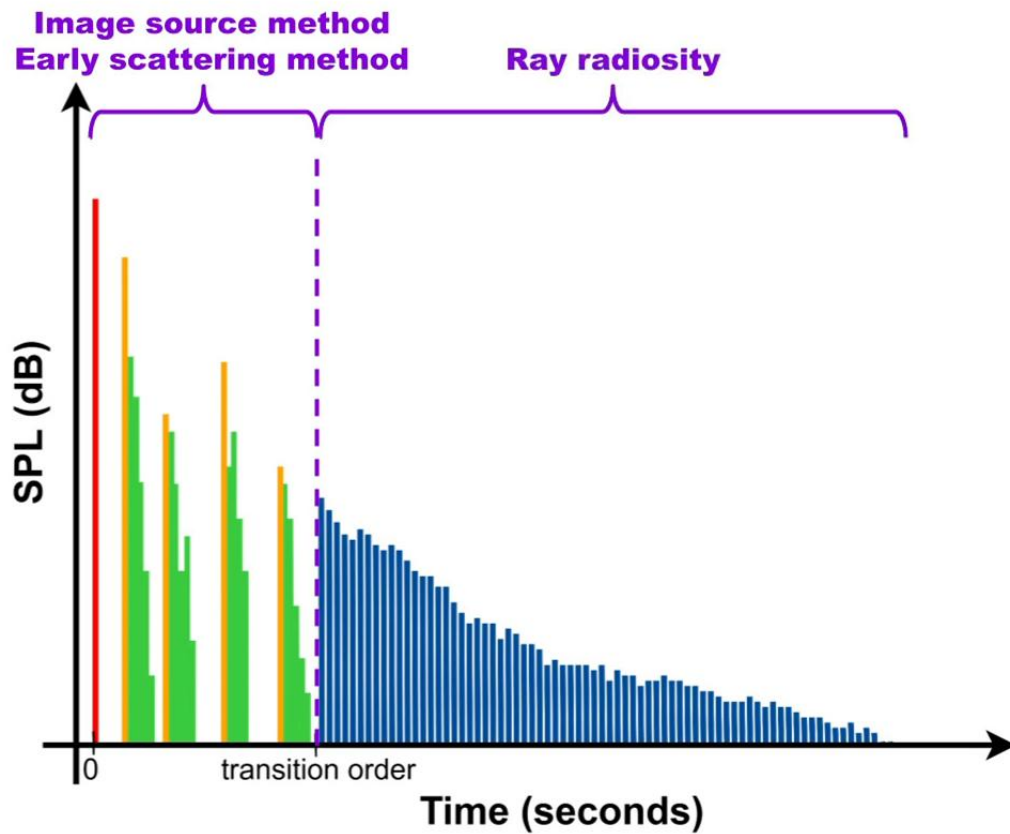
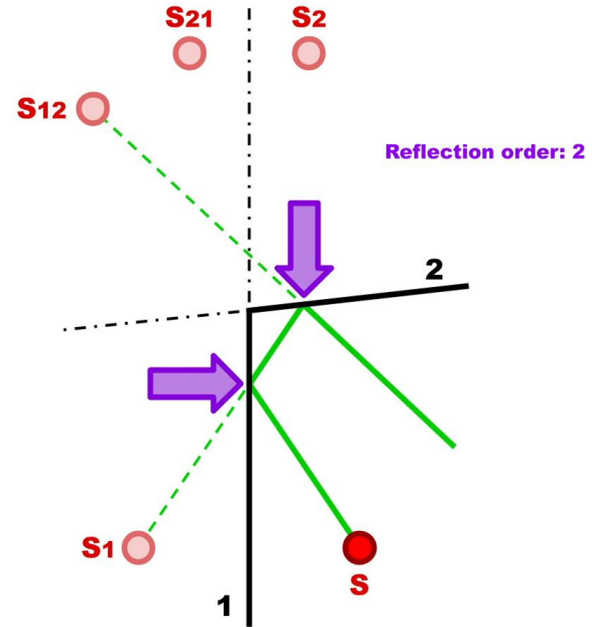
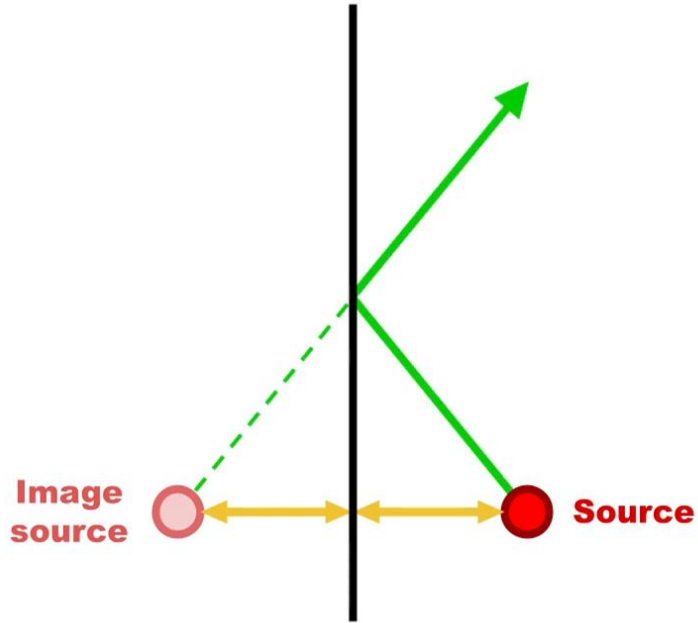
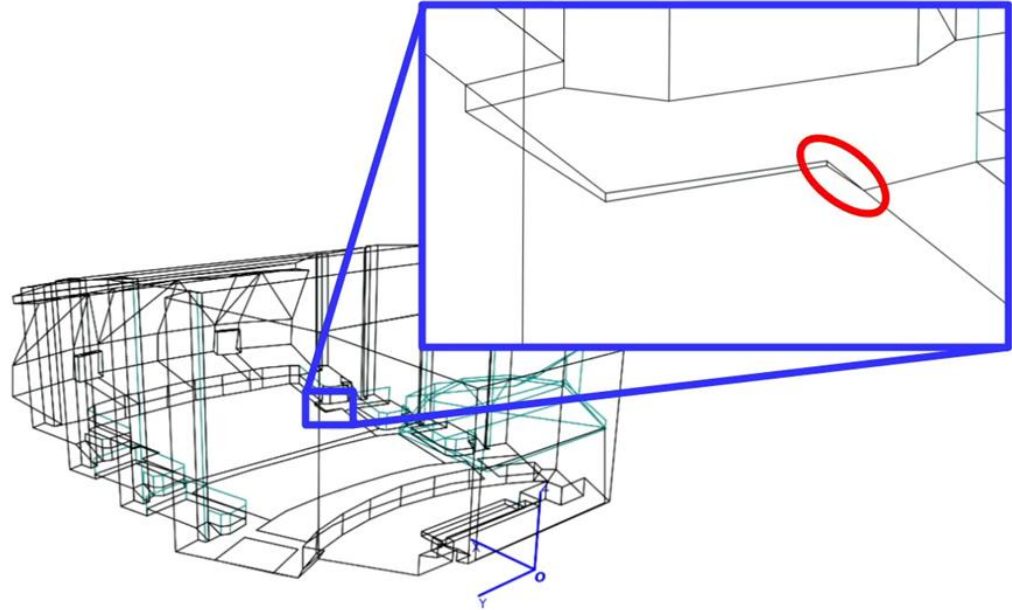
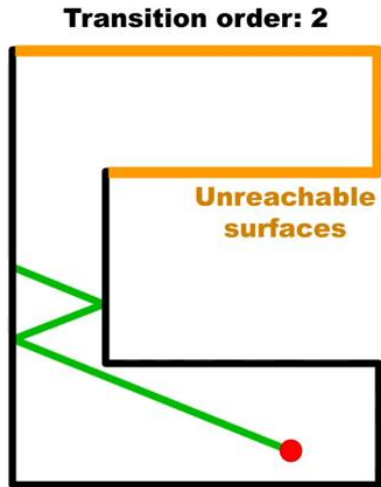
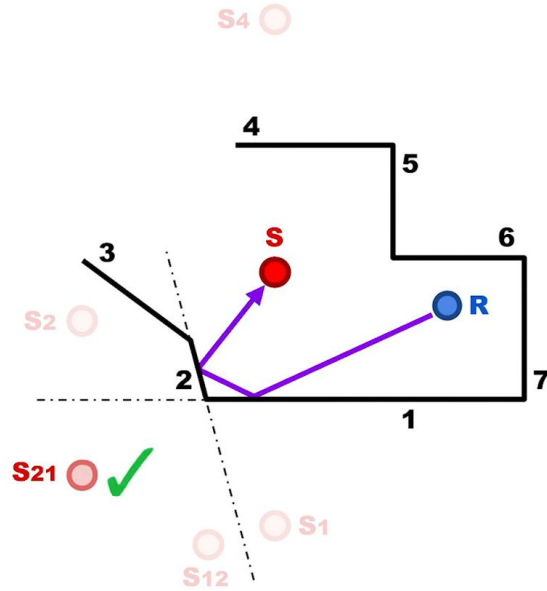


Image Source method

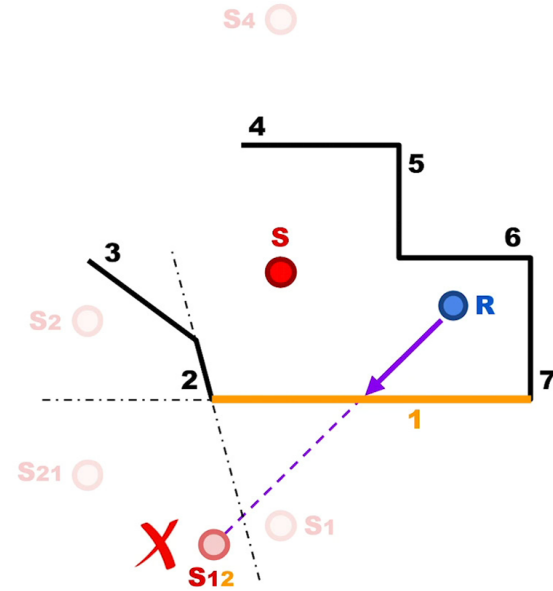


Superfici significative





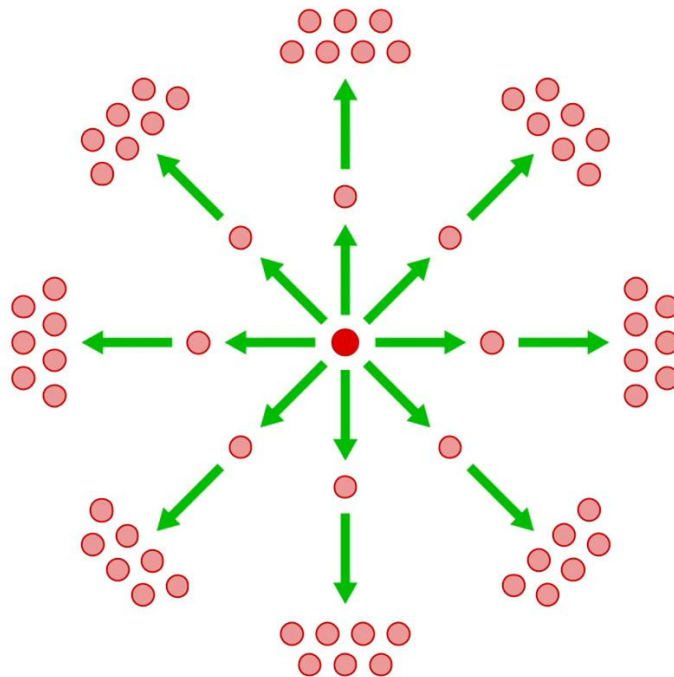
sorgente visibile



sorgente invisibile

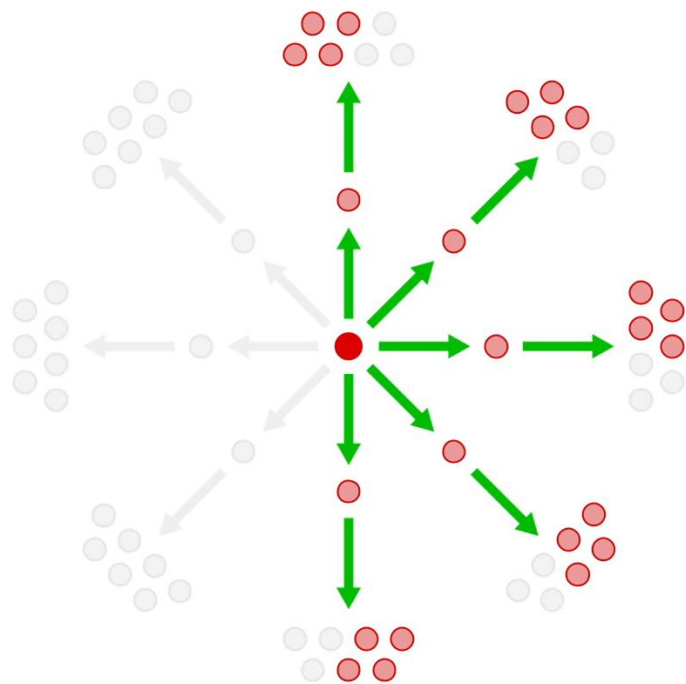
8 superfici
2 transition order

complessità alta

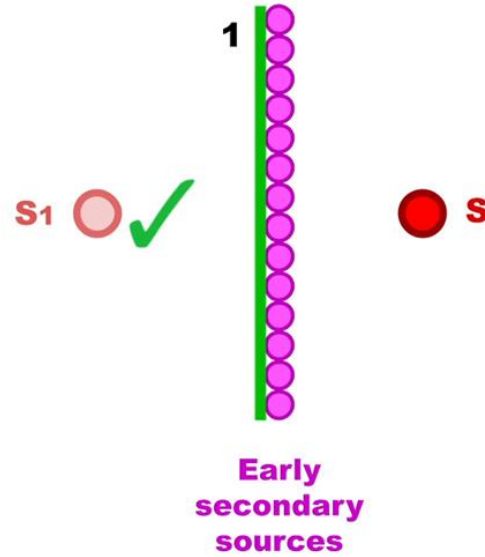


8 superfici
2 transition order

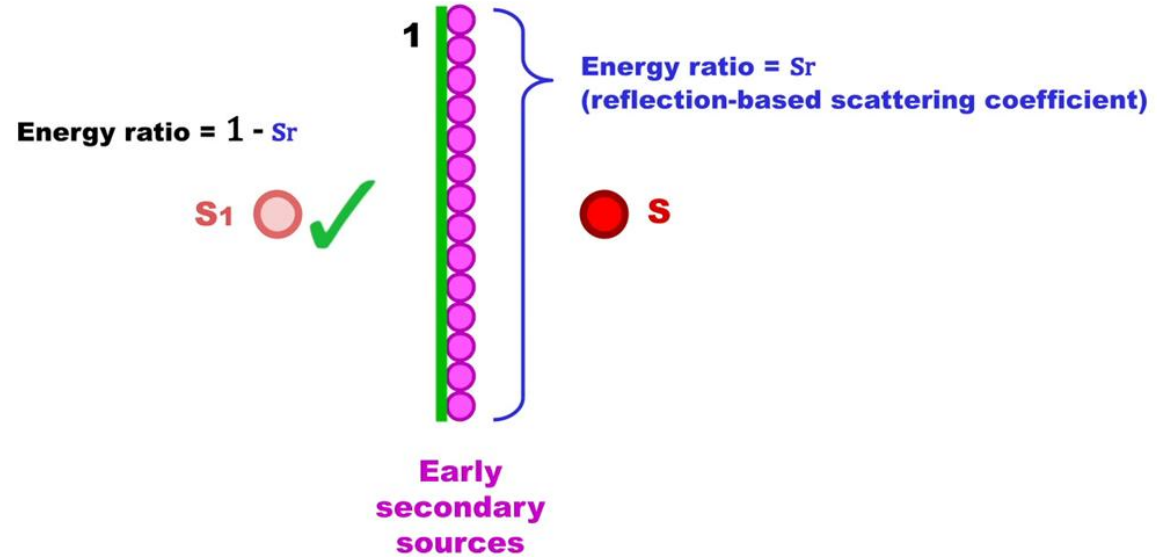
complessità bassa



Early Scattering method

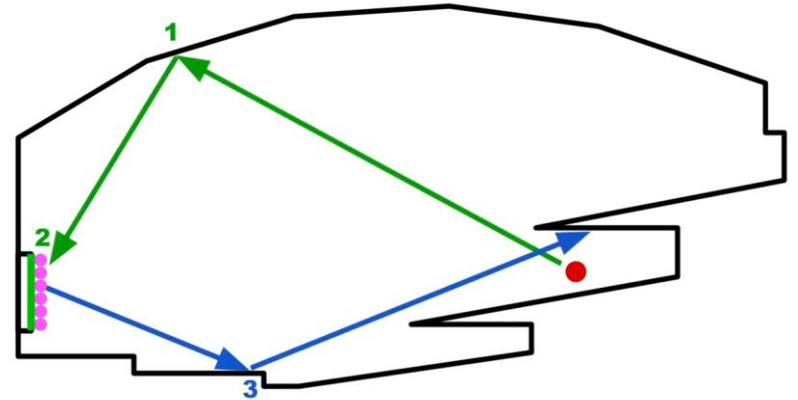
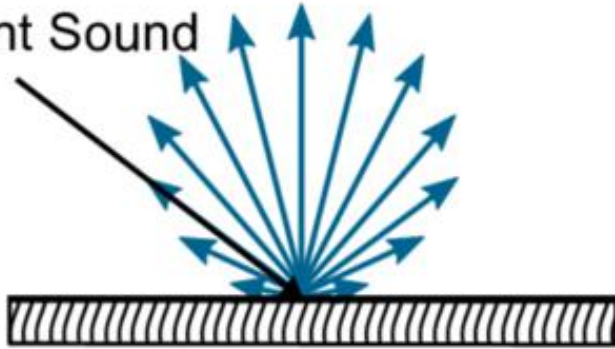


Early Scattering method

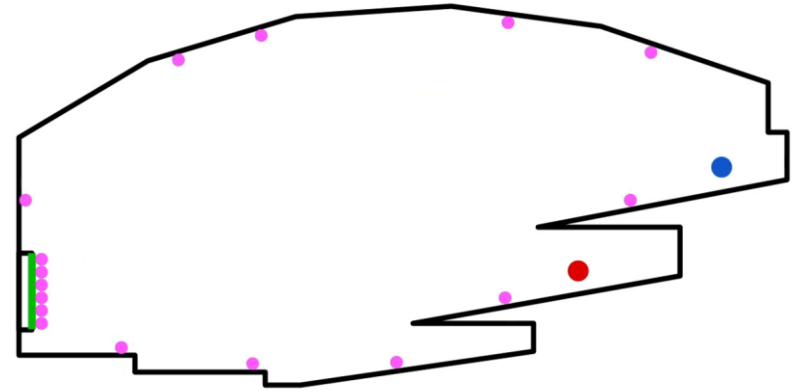
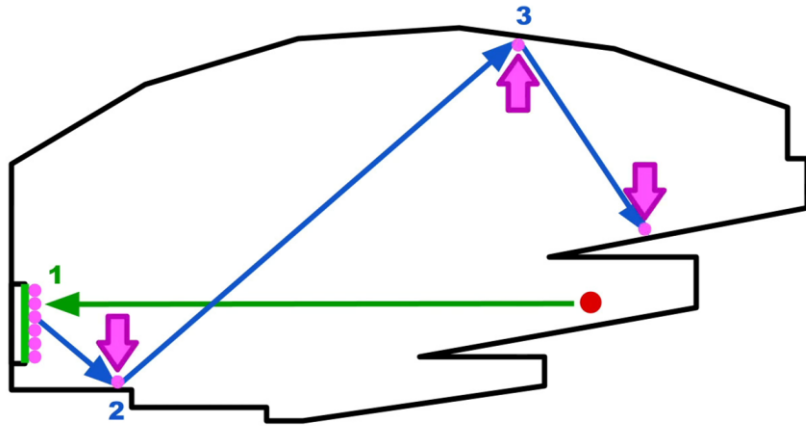


Diffusione di Lambert ed Ereditarietà

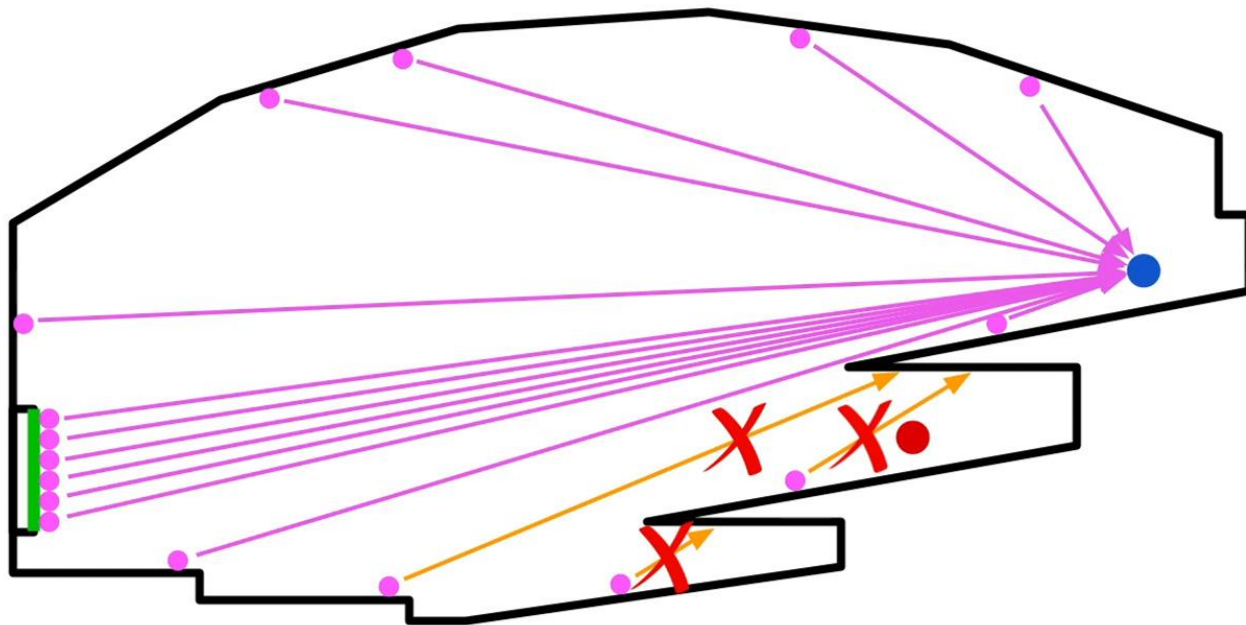
Incident Sound



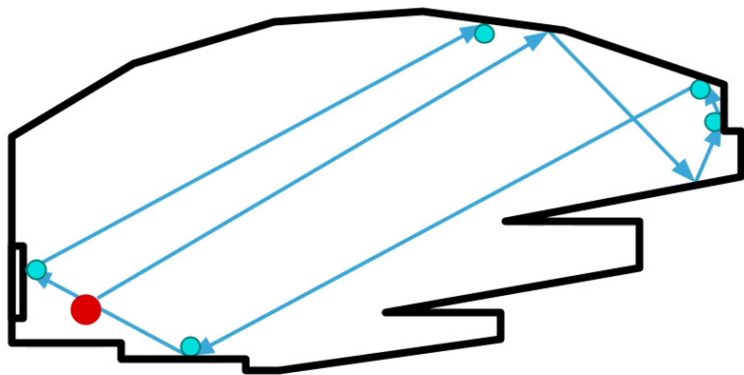
Early secondary sources



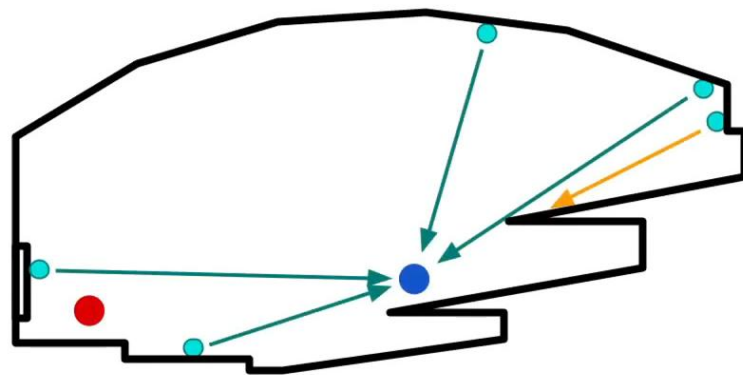
Visibilità (ESS)



Ray Radiosity

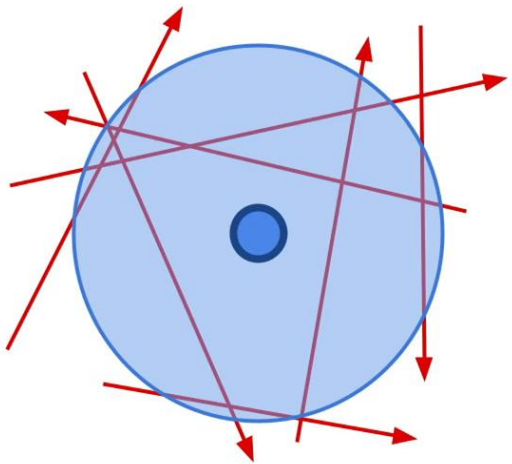


Rendering

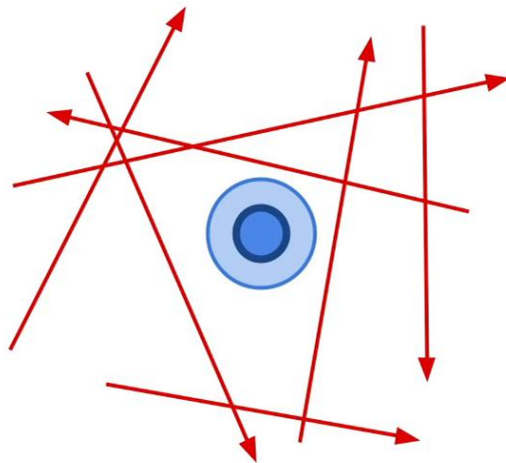


Gathering

Ray Tracing classico

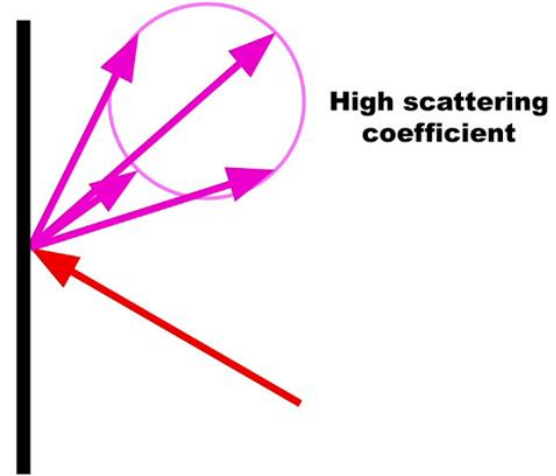
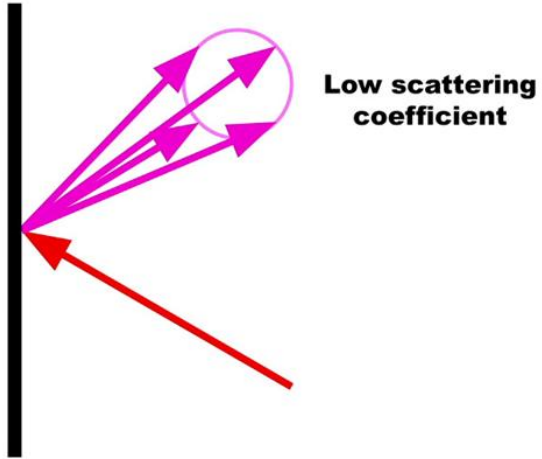


troppo grande

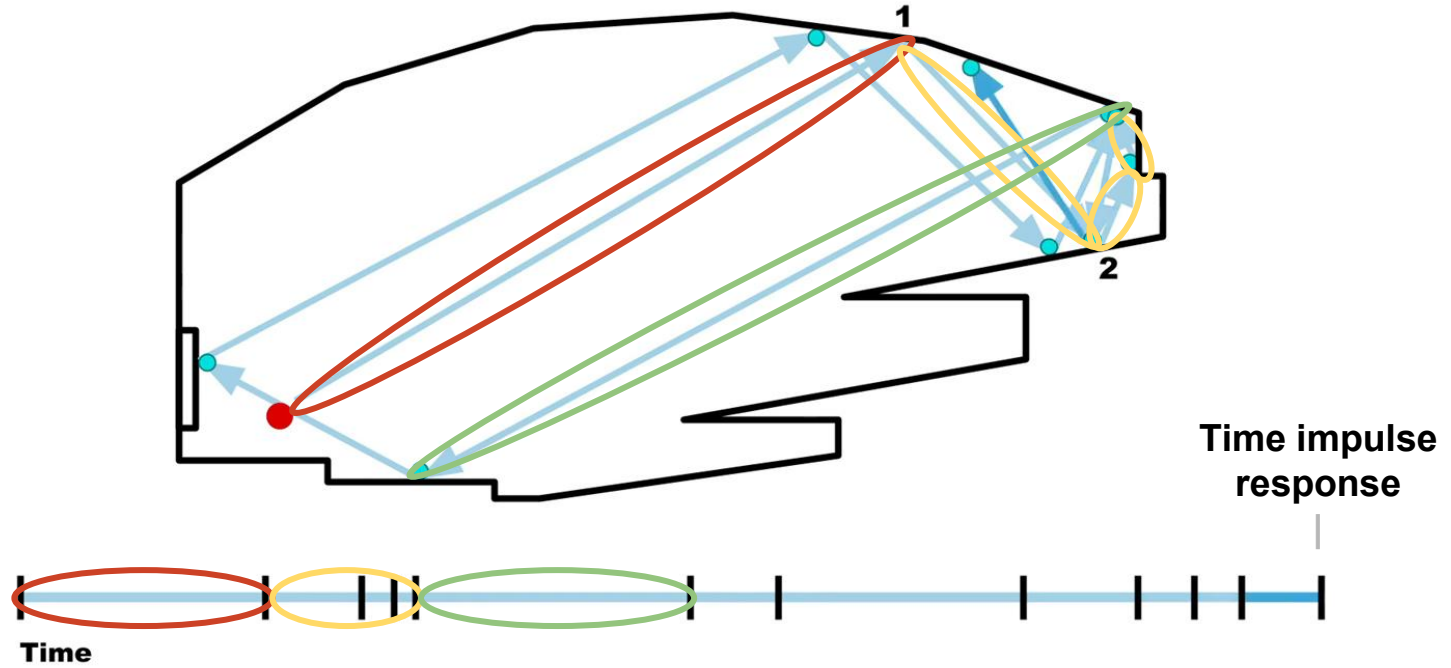


troppo piccola

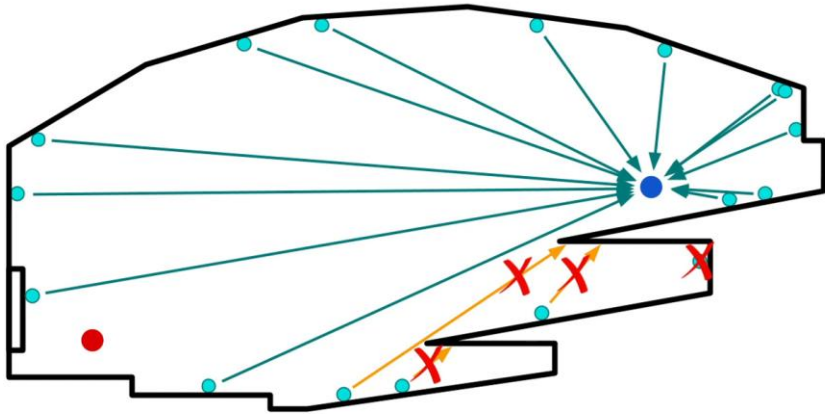
Scattering vector-based



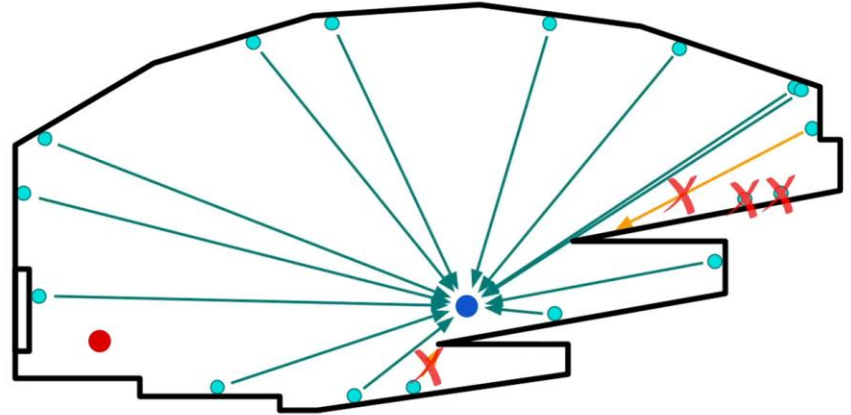
Late secondary sources



Gathering

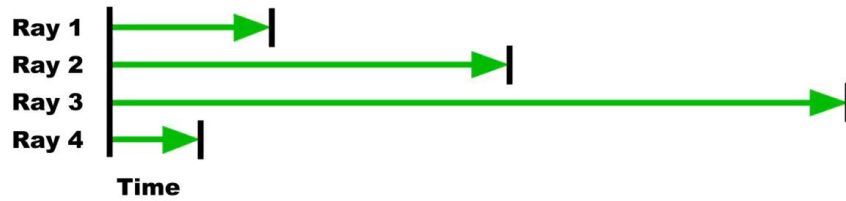
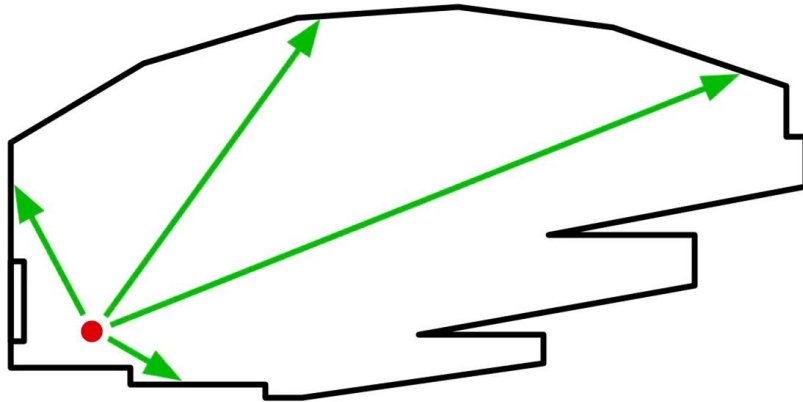


Ricevitore #1

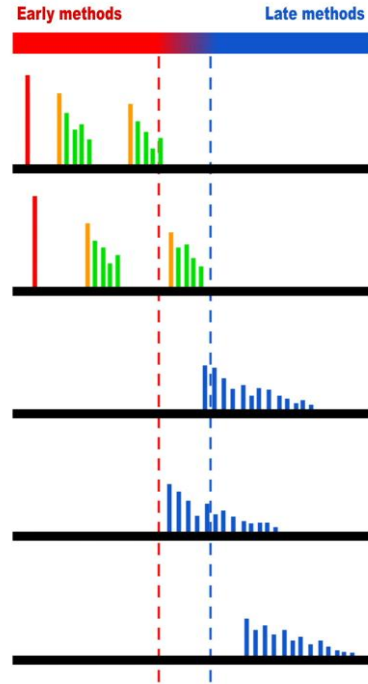
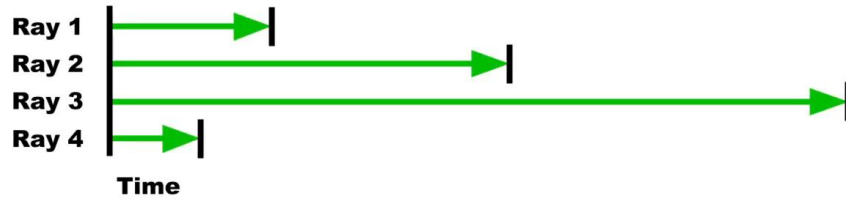
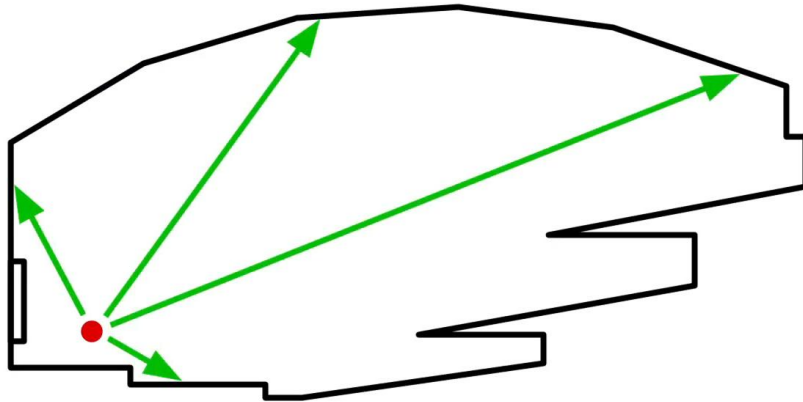


Ricevitore #2

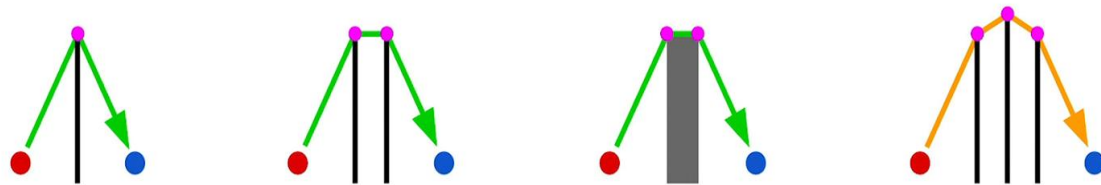
Timing



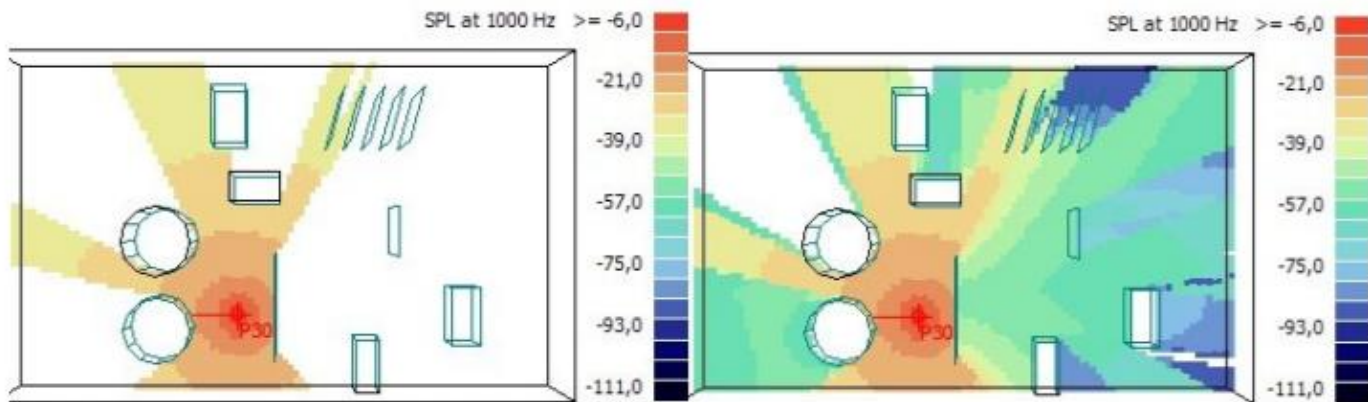
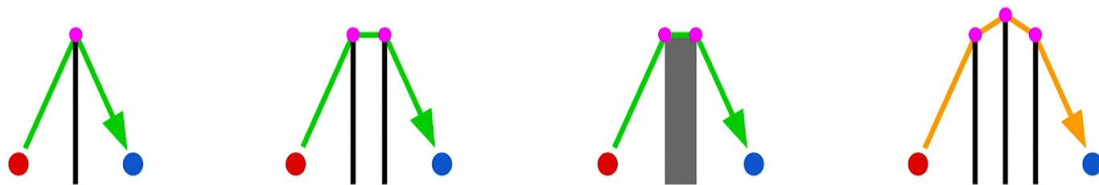
Timing



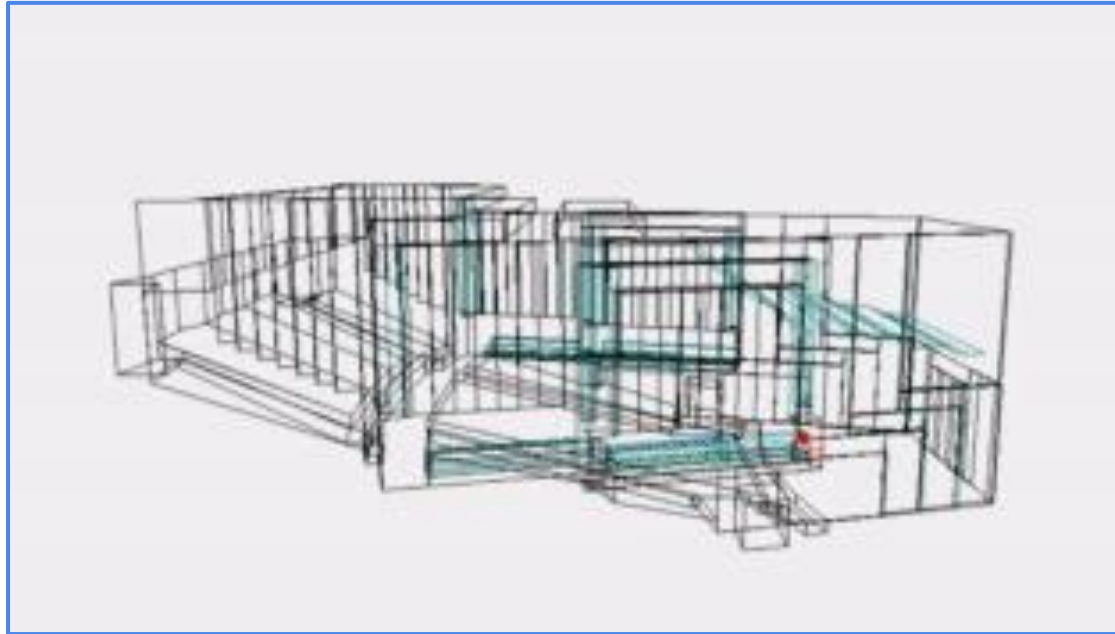
Calcolo della Diffrazione



Calcolo della Diffrazione



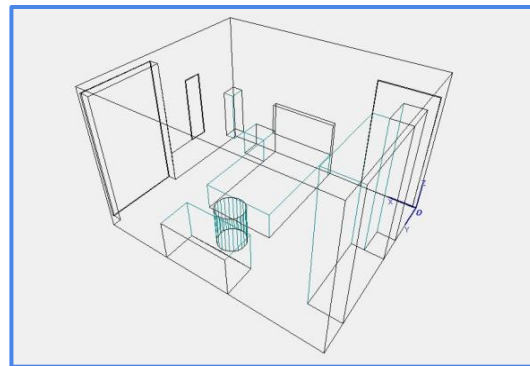
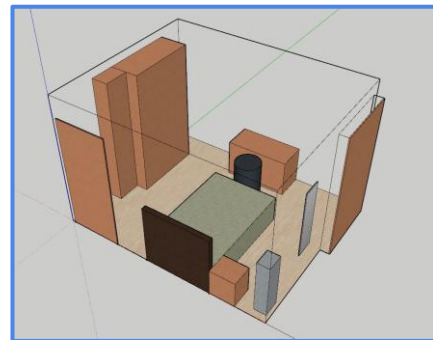
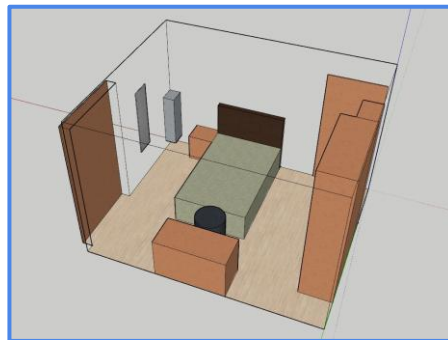
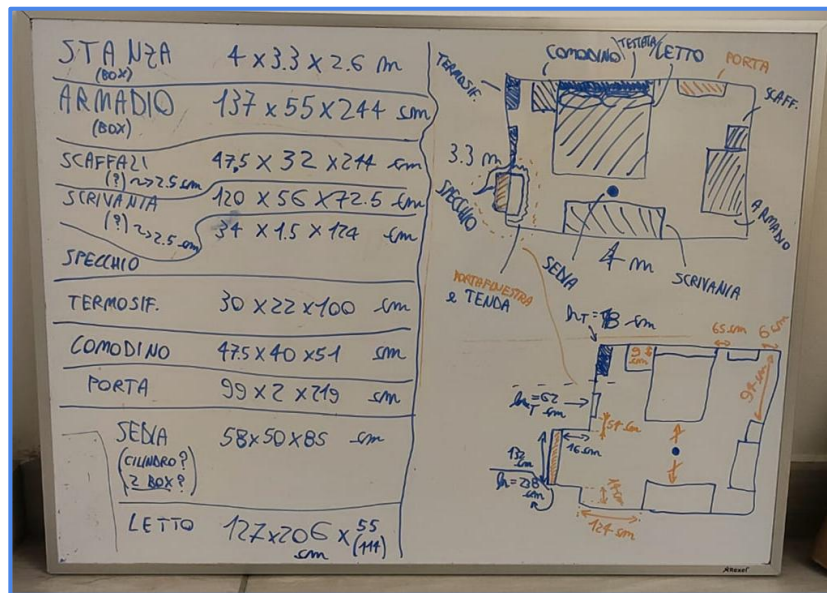
Particle Tracing



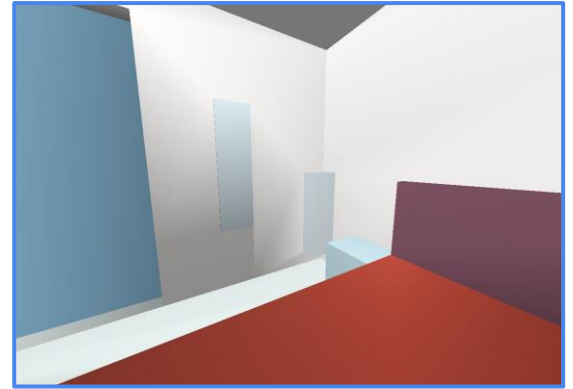
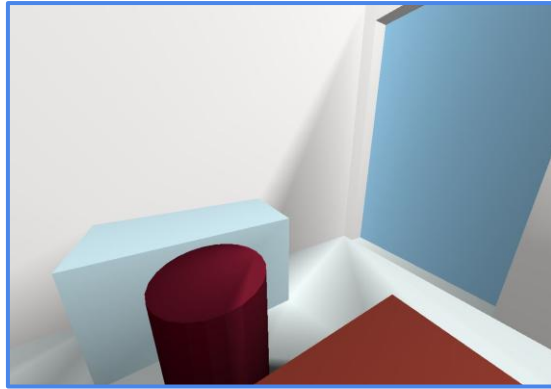
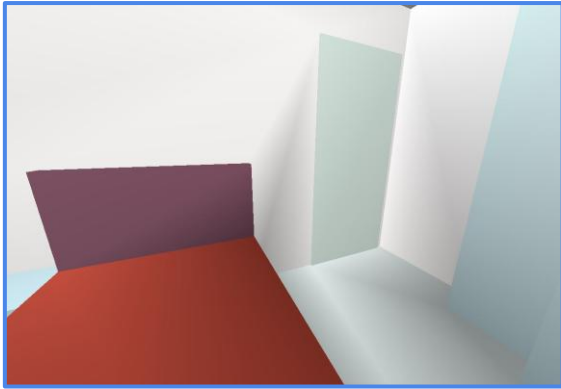
Simulazione: acustica attuale

Procedimento e parametri scelti

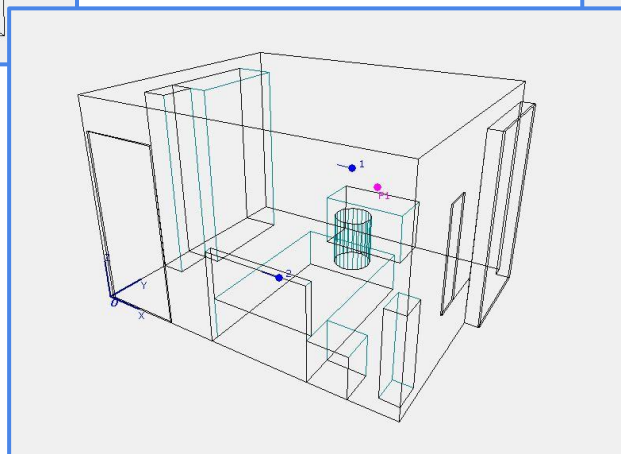
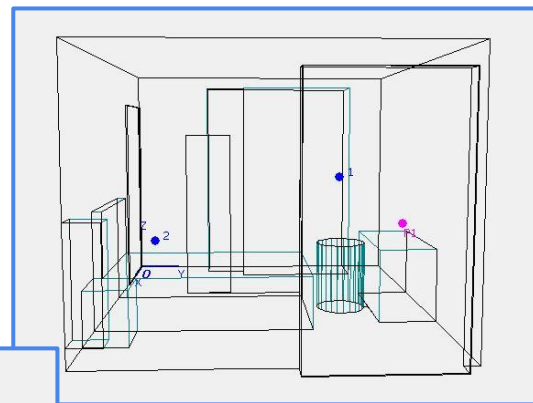
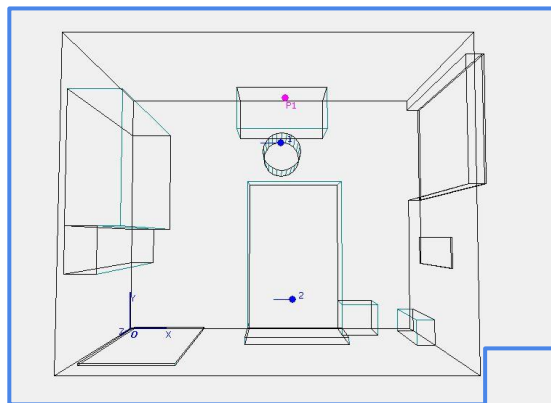
Misurazione, modellazione ed import



Assegnazione materiali



Sorgente e Ricevitori



Room setup

Calculation parameters Air conditions/STI parameters/model check

Let ODEON suggest calculation setup for point responses

Survey **Engineering** **Precision**

General settings

Impulse Response Length 1200 ms

Number of late rays (Recom. 1000) 1000

Specialist settings

Impulse response details

Max. reflection order 10000

Impulse response resolution 1.0 ms

Min. distance to walls 0.100 metres

Early reflections

Transition Order 2

☐ Manual number of early rays 8000

Number of early scatter rays (per image source) 100

Select calculation methods

Angular absorption Soft materials only

☒ Screen diffraction

Surface scattering

☐ None ($s=0$) ☒ Actual ☐ Full scatter ($S=1$)

☒ Oblique Lambert

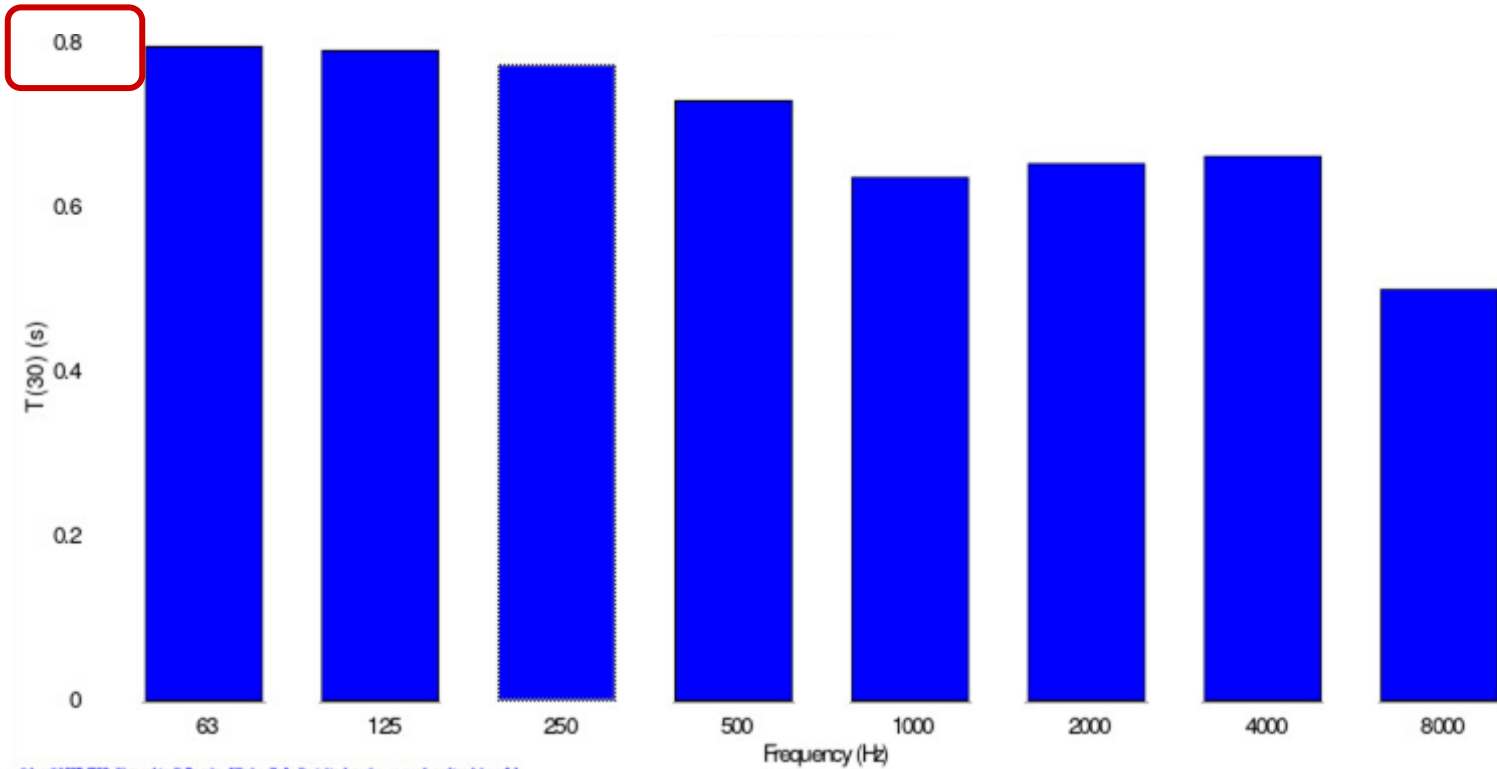
Reflection based scatter Enabled

Key diffraction frequency 707 Hz

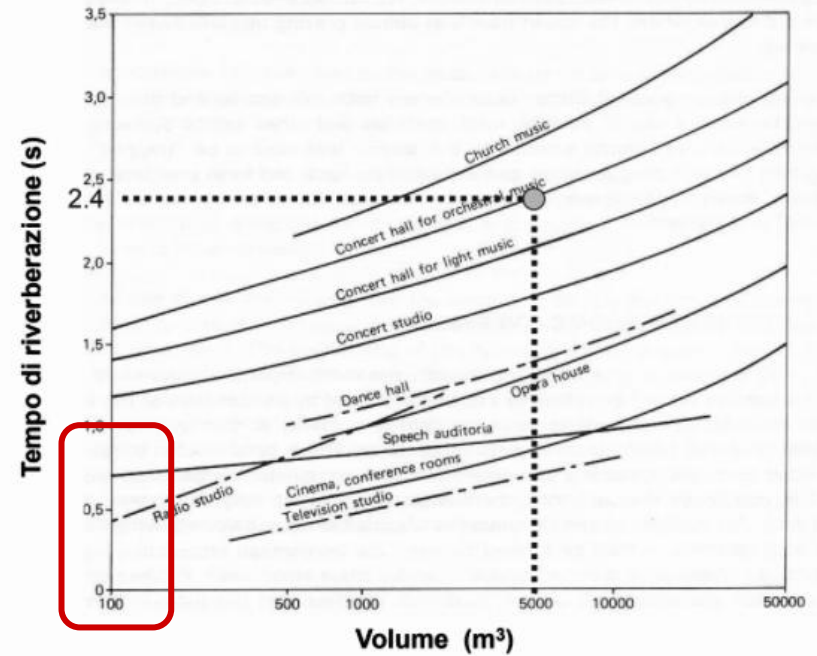
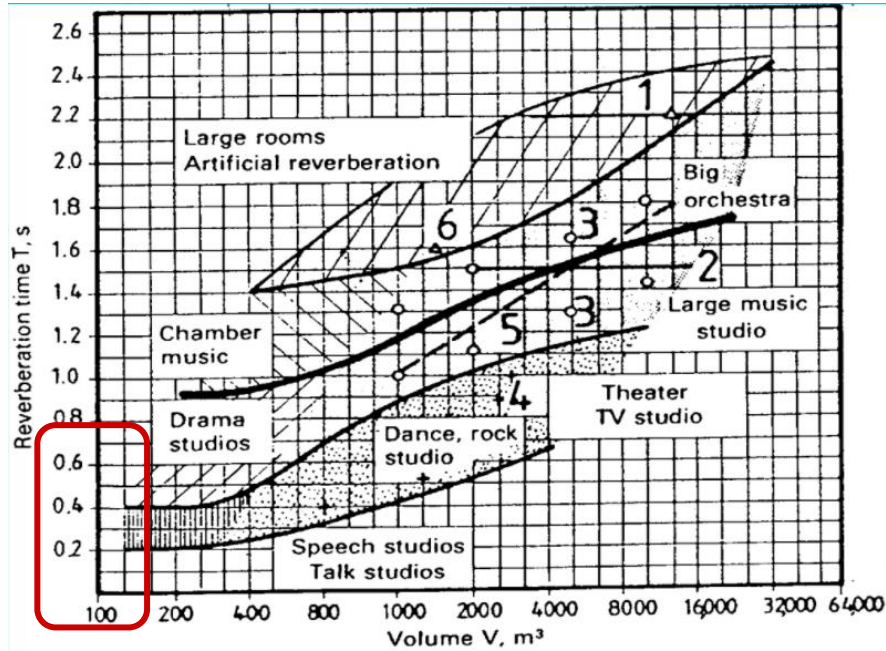
Interior margin 0.100 metres

Reflection scattering coefficients > 0.50 handled as uniform scatter (below as Lambert scatter)

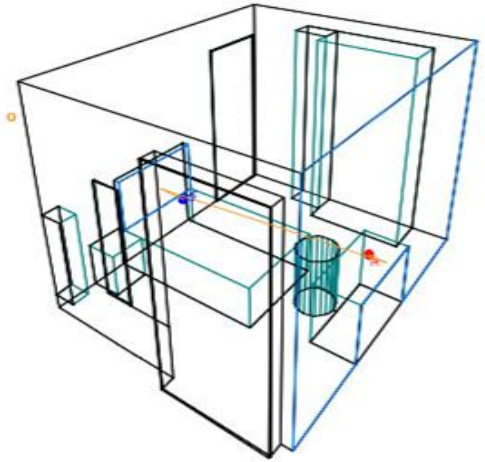
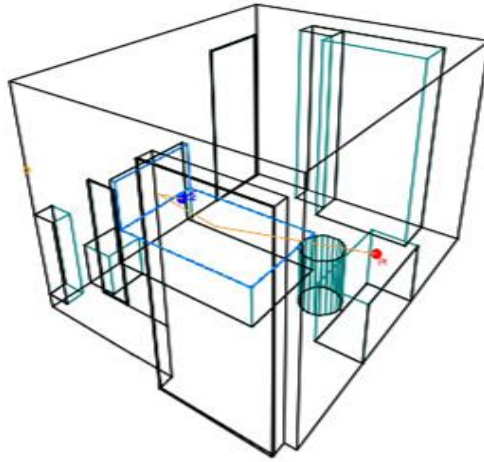
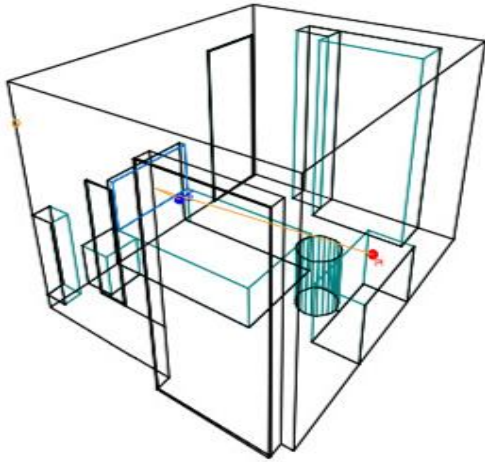
Tempo di riverbero



Tempo di riverbero ottimale



Prime riflessioni (1)



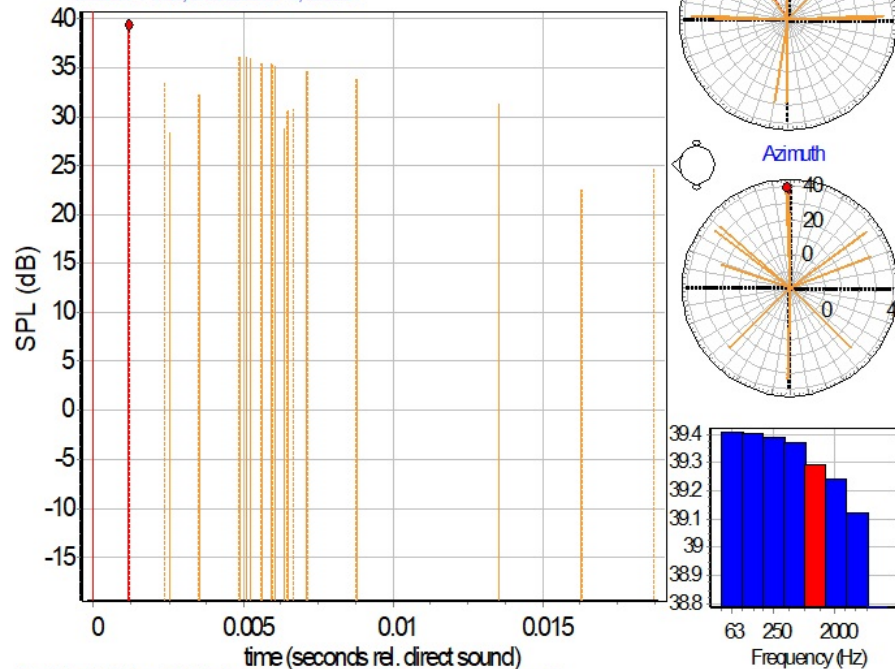
Reflectogram

Arrival time: 8.76 ms (1.16 ms rel. direct or first reflection)

Level of: 39.29 dB (-1.39 dB rel. direct)

Azimuth angle: -88.09°, elevation angle: 286°

Reflection: 1. order, 2. reflection of 19, source 1



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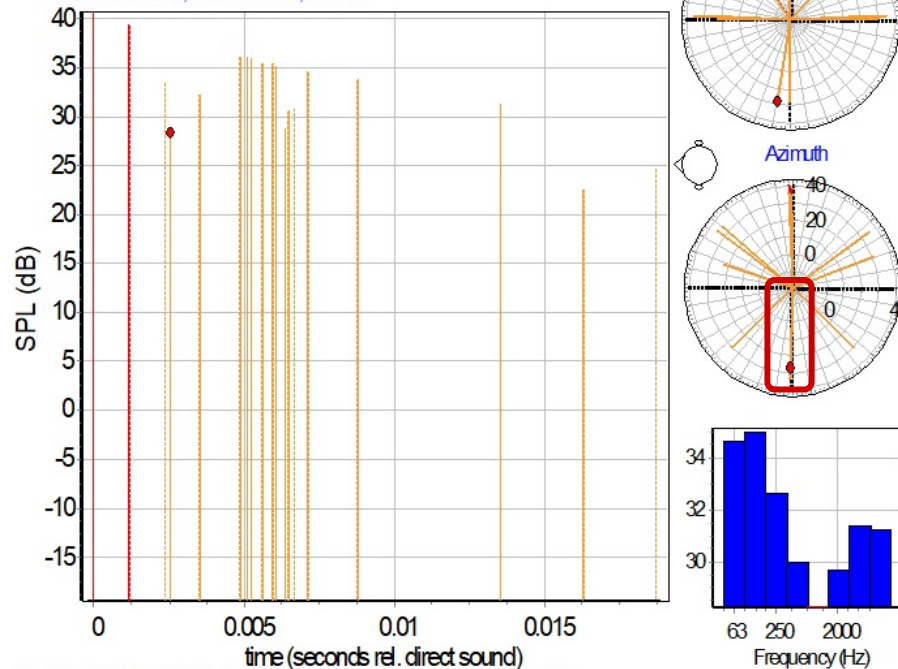
Reflectogram

Arrival time: 10.14 ms (2.54 ms rel. direct or first reflection)

Level of: 28.26 dB (-12.41 dB rel. direct)

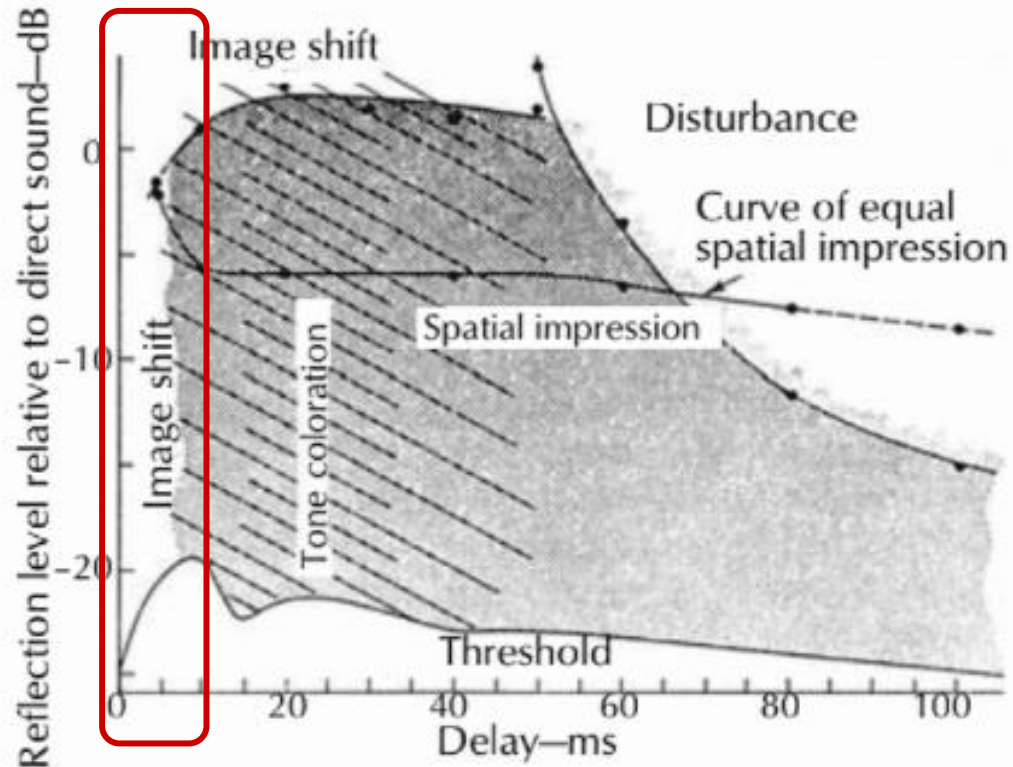
Azimuth angle: 88.33°, elevation angle: -10.43°

Reflection: 2. order, 4. reflection of 19, source 1

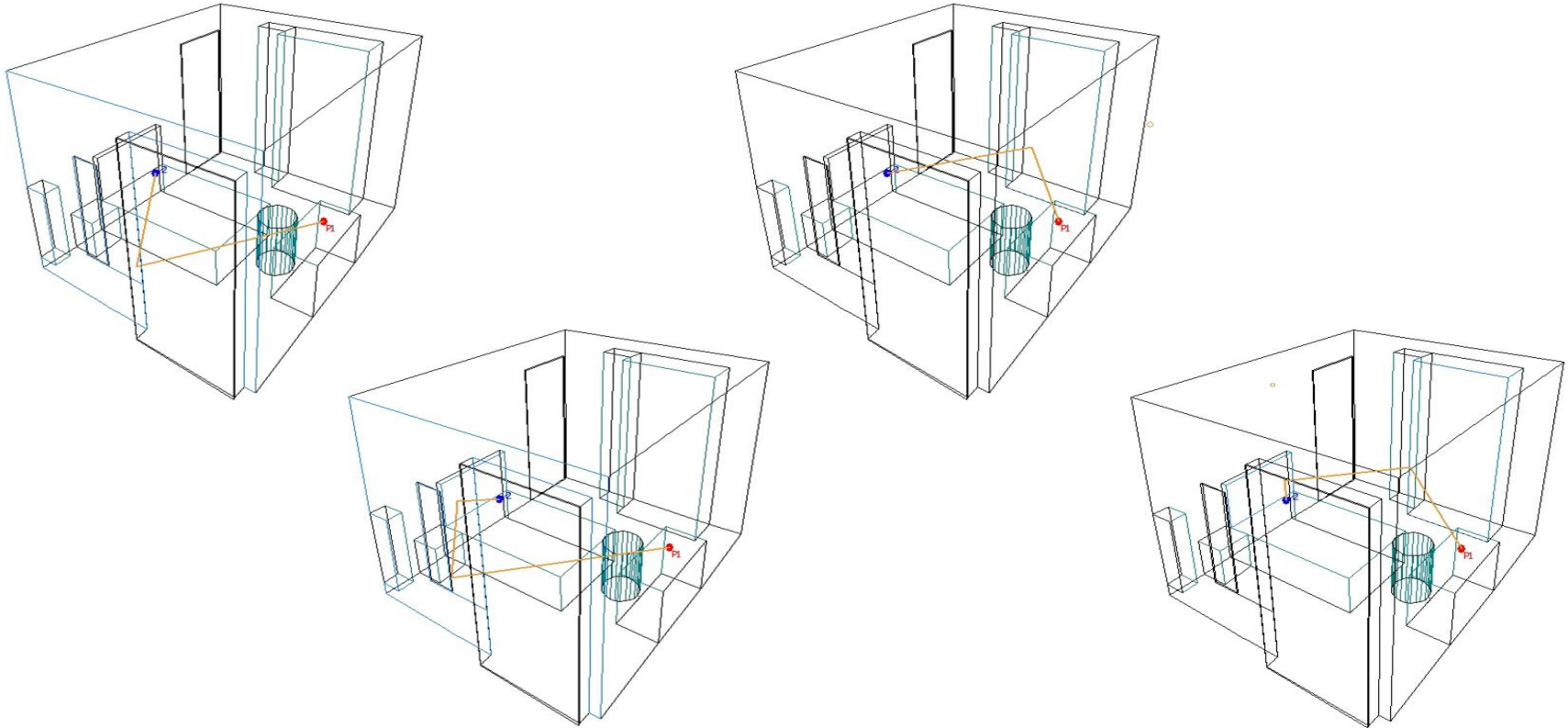


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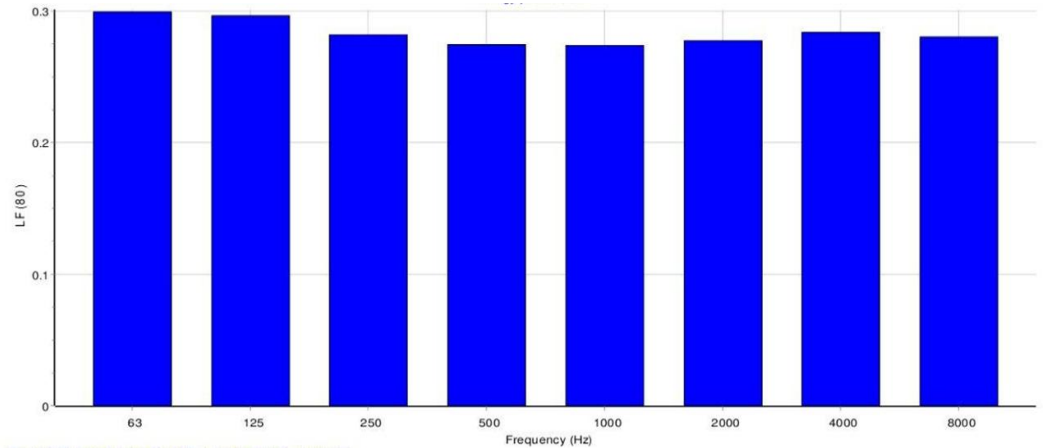
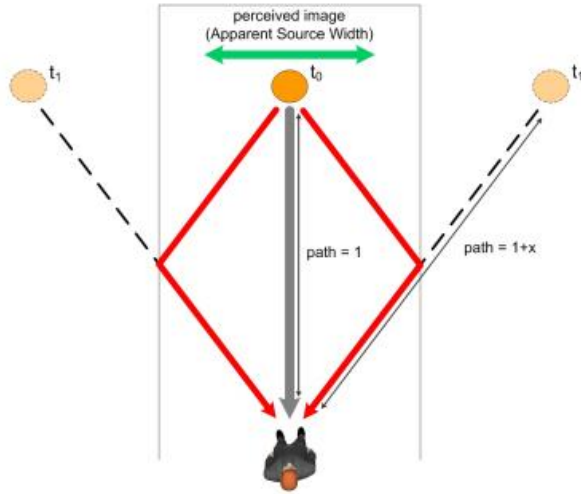
Image shift



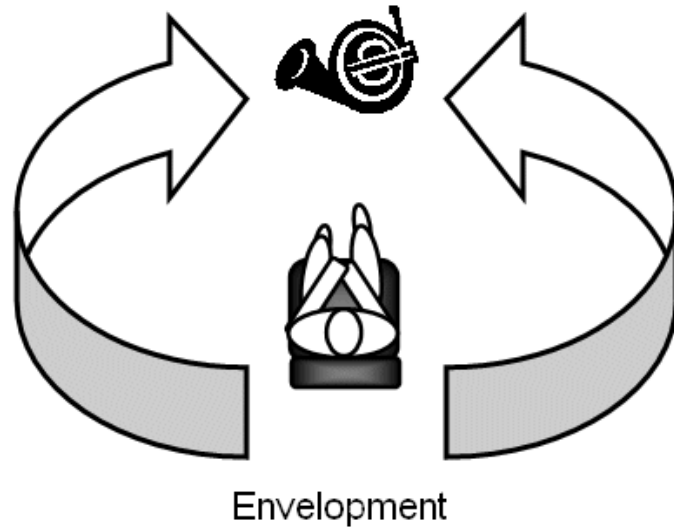
Prime riflessioni (2)



Apparent Source Width ed LF80

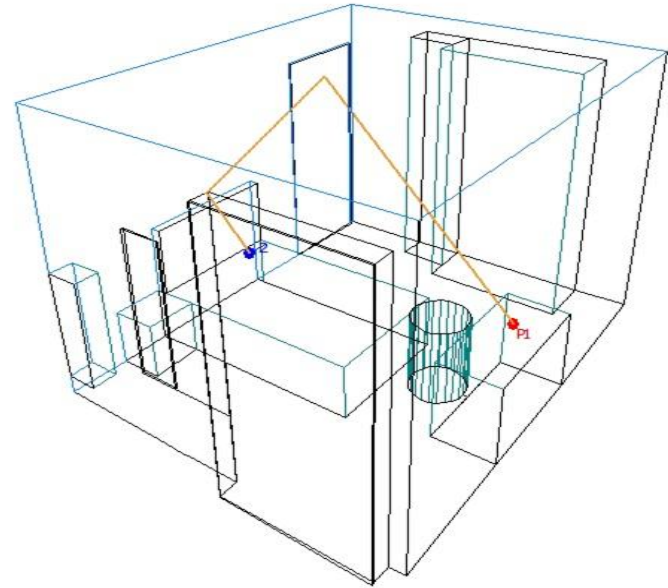
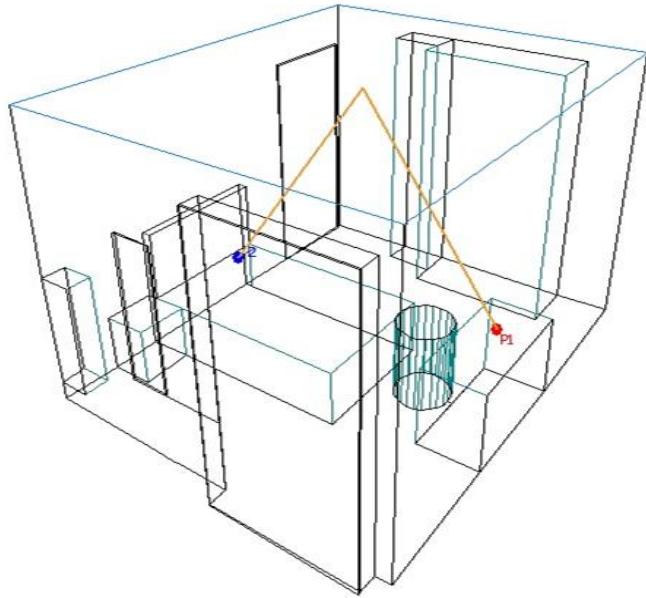


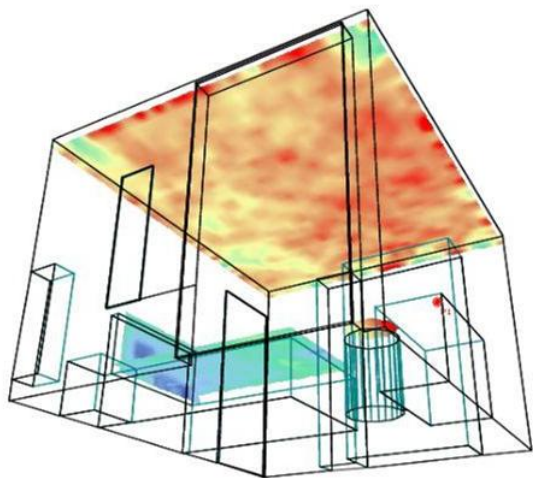
Listener Envelopment ed Lj



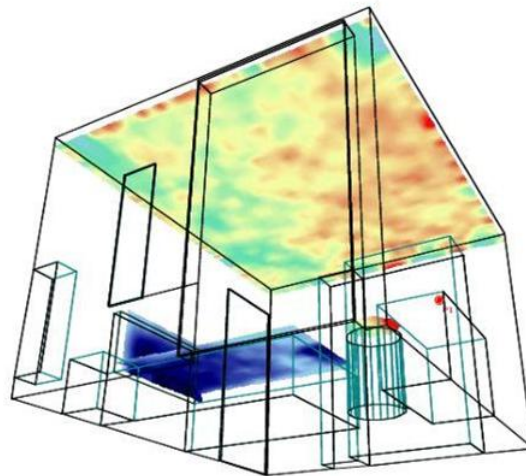
G(Average)	55.0	dB
D(50_Average)	0.66	
C(80_Average)	6.5	
Ts(Average)	47	
LF(80_Average)	0.282	
Lj(Average)	44.4	dB
BR(RT)	1.1	
BR(SPL)	1.3	dB
SIL	47.6	dB
AI	1.00	
Alcons(STI)	4.90	%

Prime riflessioni (3)





SPL a 500 Hz



SPL a 1000 Hz



Simulazione: progetto acustico

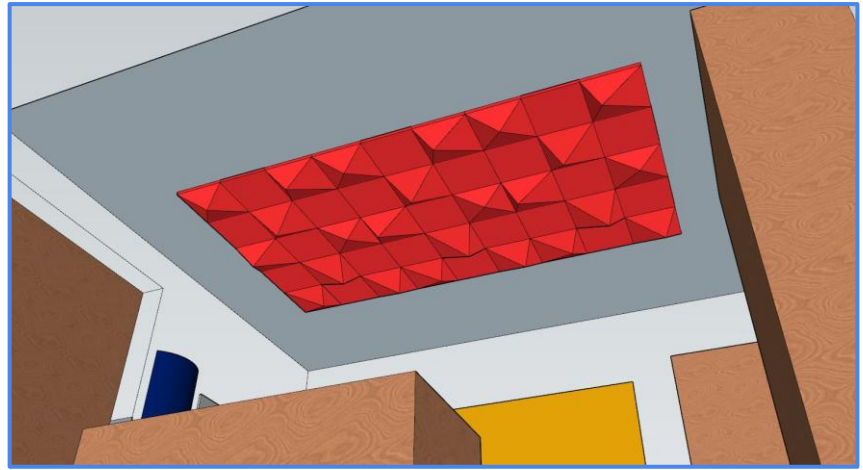
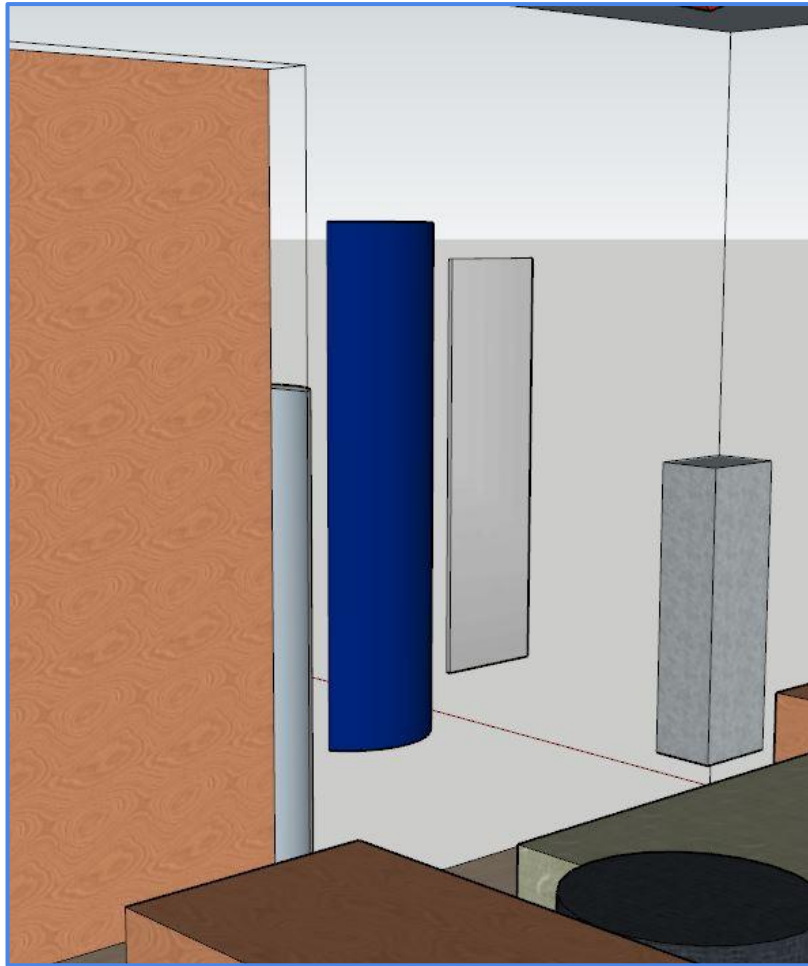
Risoluzione ai problemi e migliorie acustiche

Problemi - Soluzioni

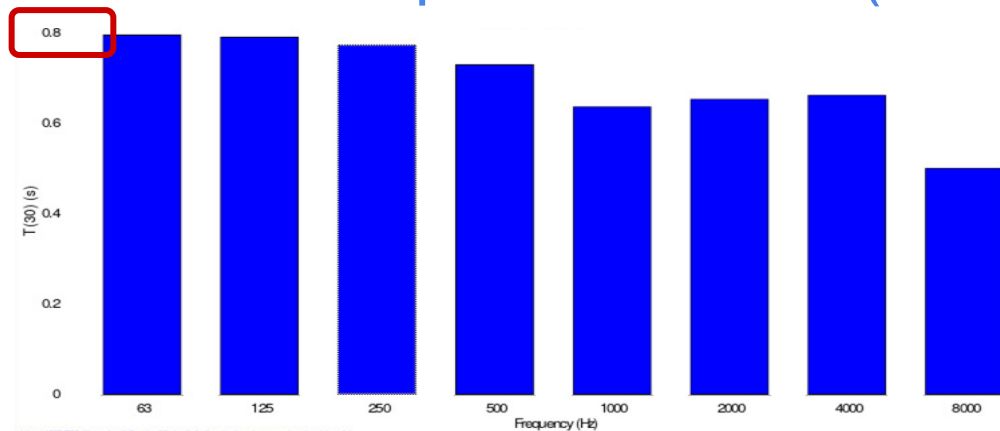
- Tempo di riverbero
- Riflessioni da dietro
- Riflessioni laterali e diffusione
- Riflessioni dall'alto



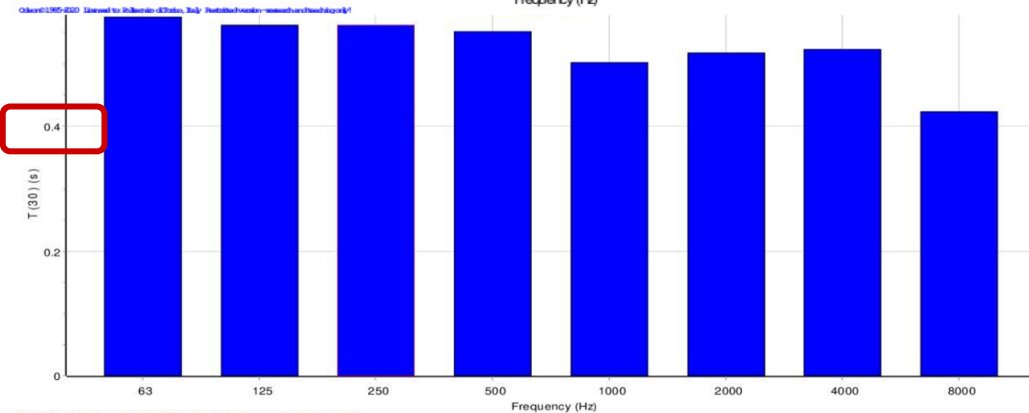
- Pannelli fonodiffondenti laterale e sul soffitto
- Pannelli fonoassorbenti sul soffitto e sulla parete dietro
- Aumento assorbenza della testata del letto



Tempo di riverbero (confronto)

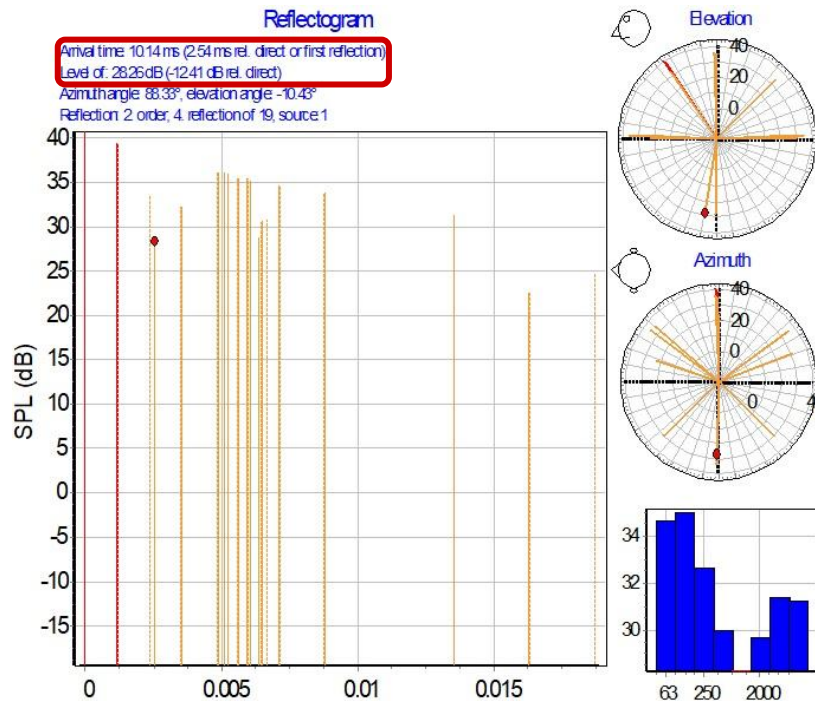


Attuale

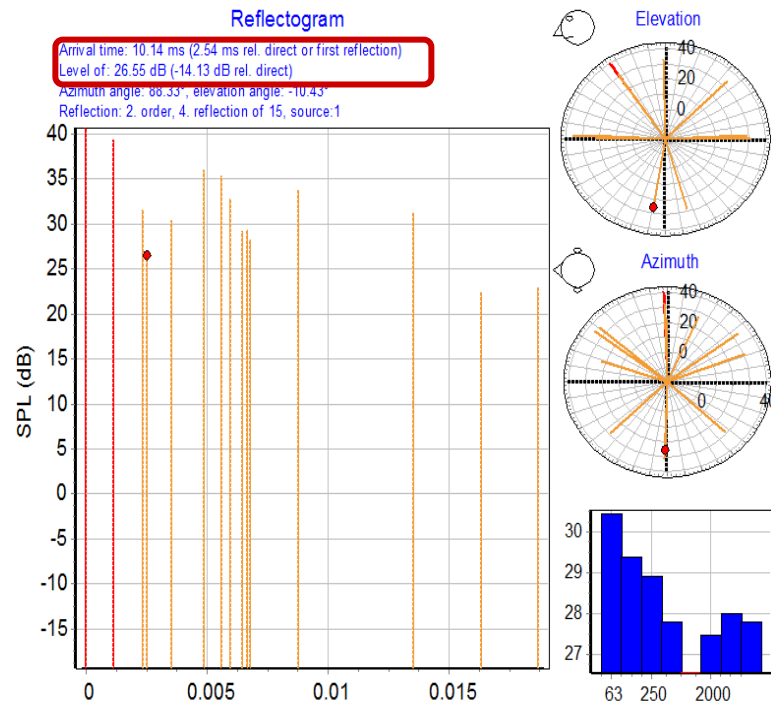


Progetto

Riflessioni da dietro (confronto)

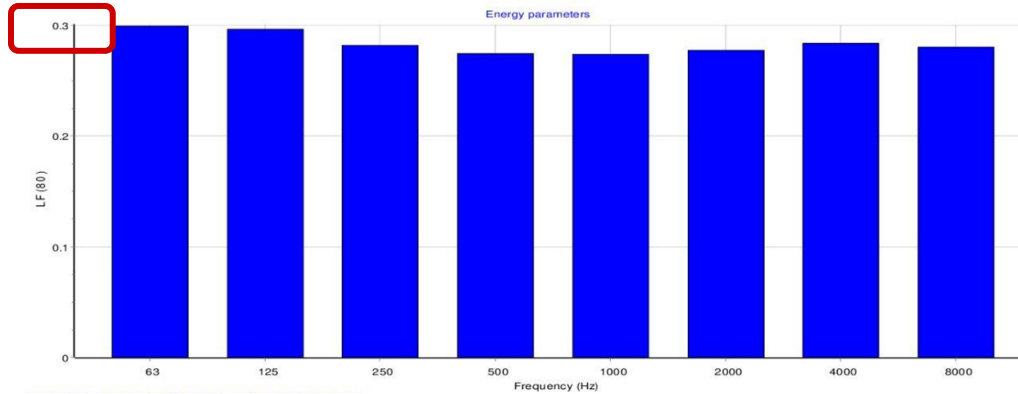


Attuale

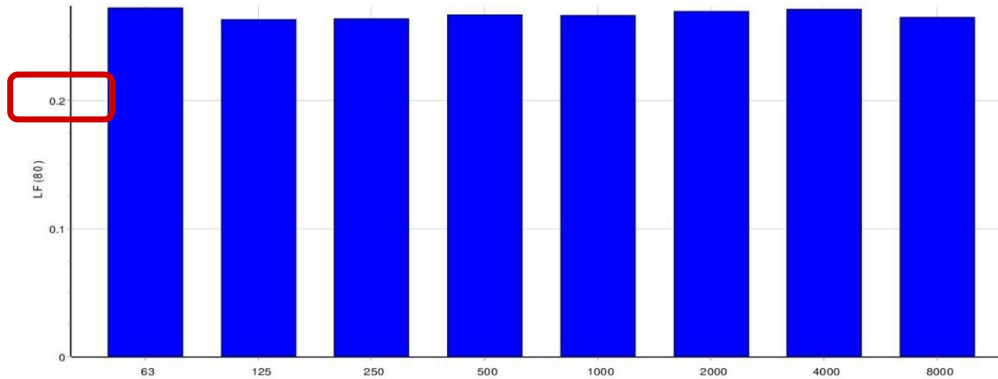


Progetto

ASW ed LF80 (confronto)



Attuale



Progetto

LEV ed Lj (confronto)

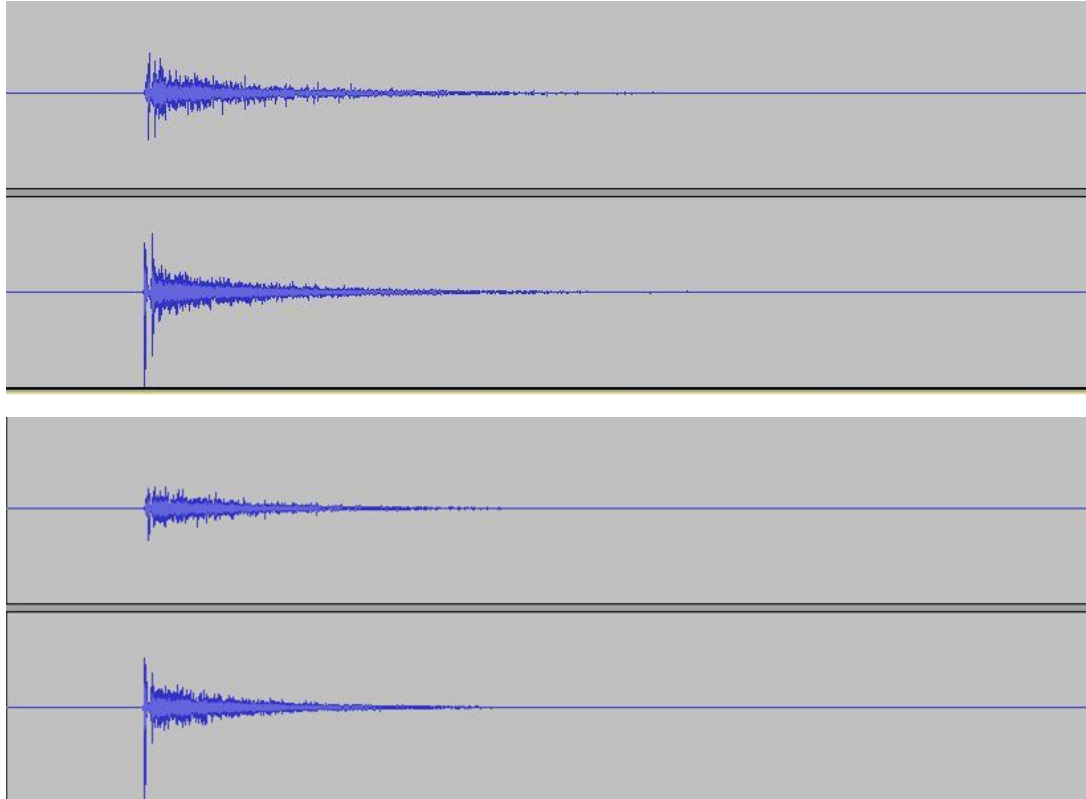
G(Average)	55.0	dB
D(50_Average)	0.66	
C(80_Average)	6.5	
Ts(Average)	47	
LF(80_Average)	0.282	
Lj(Average)	44.4	dB
BR(RT)	1.1	
BR(SPL)	1.3	dB
SIL	47.6	dB
AI	1.00	
Alcons(STI)	4.90	%

Attuale

G(Average)	55.7	dB
D(50_Average)	0.72	
C(80_Average)	8.3	
Ts(Average)	39	
LF(80_Average)	0.265	
Lj(Average)	40.4	dB
BR(RT)	1.1	
BR(SPL)	0.8	dB
SIL	47.6	dB
AI	1.00	
Alcons(STI)	3.98	%

Progetto

Risposta all'impulso (confronto)



$x(t)$



$h(t)$

$y(t)$

Attuale



$h(t)$

$y(t)$

Progetto



Bibliografia e Sitografia

- “Handbook for Sound Engineers”, Glen Ballou
- ODEON user manual, ODEON A/S
- [ODEON learn site](#)
- Dispense del corso, Masoero, Shtrepi, Fringuellino, Prato

FINE

Grazie per l'attenzione!